

Choosing dimensionality-reduction methods from evidence, not preference

Y. Gao · evidence frozen 2026-09-29 (commit 766f1d17e2, 1,751 checksummed files) · tables and figures rendered from the frozen artifacts; narrative drafted with AI coding assistance and reviewed by the author

Summary. Agent 1 is a lightweight, single-orchestrator agent for numerical data. It inspects the data, chooses preprocessing from the inspected profile, and pilots all ten supported dimensionality-reduction methods on one common cohort. Only methods supported by label-free evidence advance. On PBMC3k it retained PCA, MDS, t-SNE and analysed all 2,638 post-QC cells. On PathMNIST it retained PCA, MDS, t-SNE, UMAP and analysed a stratified subset of 12,500 of the 89,996 training images; measured scaling set that size. A first version (v1) produced plausible reports, but an audit showed that its shortlist was partly decided by import-time noise and by a stability score that was not comparable across methods. The version reported here (v2) corrects the measurements, fixes the decision rules before the real-data run, and reports how robust each decision is.

1 · Problem and contribution

A dimensionality-reduction workflow should explain why a method advances on the dataset at hand. Rather than selecting methods solely from hard-coded heuristics or LLM recommendations, the agent runs a lightweight pilot benchmark across all supported methods. It evaluates their empirical behavior quantitatively and uses that evidence to decide which methods advance to full-scale analysis.

v2 adds three things to that principle:

- **Comparable evidence:** fit-only timing, stability measured the same way for every method, local and global structure metrics, and explicit failure diagnostics.
- **Pre-registered decisions:** weights and thresholds were committed before the real-data run, and the robustness of every decision to reasonable rule changes is reported.
- **Evidence-based final size:** the final cohort size is derived from measured scaling of the shortlisted methods, with no dataset-specific branch.

2 · Architecture

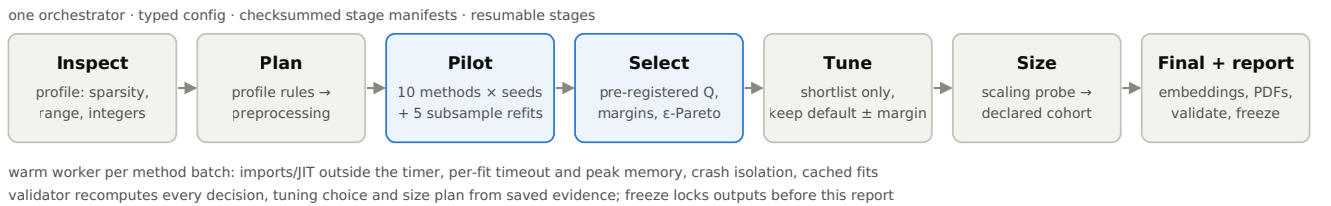


Figure 1. Stages and the deterministic tools behind them. Every arrow passes saved, checksummed artifacts; any stage can be resumed or recomputed on its own.

A Python package exposes the stages through one CLI; SLURM runs every heavy step. Scikit-learn, pydiffmap, GPyTorch and umap-learn supply the methods; Jinja2 and WeasyPrint render PDFs, checked with pypdf. There is no multi-agent framework, database or web service. Labels only stratify cohorts and color secondary plots.

3 · Dataset-aware planning and decision logic

Profile evidence	PBMC3k	PathMNIST
Source size	2,700 × 32,738	89,996 × 2,352
Sparsity · non-negative · integer · max	0.974 · True · True · 419	0.000 · True · True · 255
Branch chosen by rules	count matrix	bounded intensity
Executed steps (per cohort)	observation qc → sample cohort → filter features → normalize log1p → select highly variable features → scale clip → common pca representation	sample cohort → scale intensity → common pca representation

Evidence. All ten methods receive the identical input: 1,000 observations and the first 50 principal components of the preprocessed features. Stochastic methods run with 3 seeds, and every method is refitted on the same 5 random 80% subsamples. Metrics are label-free and Euclidean in that input space:

- **T** (trustworthiness) and **R** (15-nearest-neighbor recall) measure local structure.
- **G** is the Spearman correlation of pairwise distances, a global measure.
- **S** is the kNN overlap between subsample refits, a stability measure that applies to deterministic and stochastic methods alike.

Q = 0.35 T + 0.35 R + 0.20 G + 0.10 S. A method is ineligible after a failure or a severe diagnostic: a disconnected or singular graph, a collapsed layout, a near-disconnected diffusion graph, or SMACOF reaching its iteration limit.

Rules. PCA is kept as the linear baseline. Among the rest, a method is eligible if any of these holds:

- it reaches 90% of the best Q;
- it is within a practical-equivalence margin of the best (max of 2 paired subsample SE and 0.005);
- it lies on a quality-runtime frontier where runtime differences count only when they are at least 2× and at least 1 s.

At most four methods advance, ranked by Q. Tuning, on the shortlist only, keeps the default unless another grid point beats it by more than the margin. The final size is the largest candidate whose conservatively projected cost fits a 30-minute per-run limit, a 40-minute stage budget and 24 GB of memory.

Method	Implementation (pilot defaults)	Method	Implementation (pilot defaults)
PCA	sklearn PCA (full SVD)	Laplacian	sklearn SpectralEmbedding (kNN 30)
Kernel PCA	sklearn KernelPCA (RBF, median γ)	Diffusion maps	pydiffmap (α 0.5, k 64, median ϵ)
MDS	sklearn metric SMACOF, classical start	GPLVM	GPyTorch Bayesian GPLVM (75 epochs)
Isomap	sklearn Isomap (k 15)	t-SNE	sklearn TSNE (perplexity 30, PCA init)
LLE	sklearn LLE (k 20, reg 1e-3)	UMAP	umap-learn (k 15, min_dist 0.1)

4 • Results

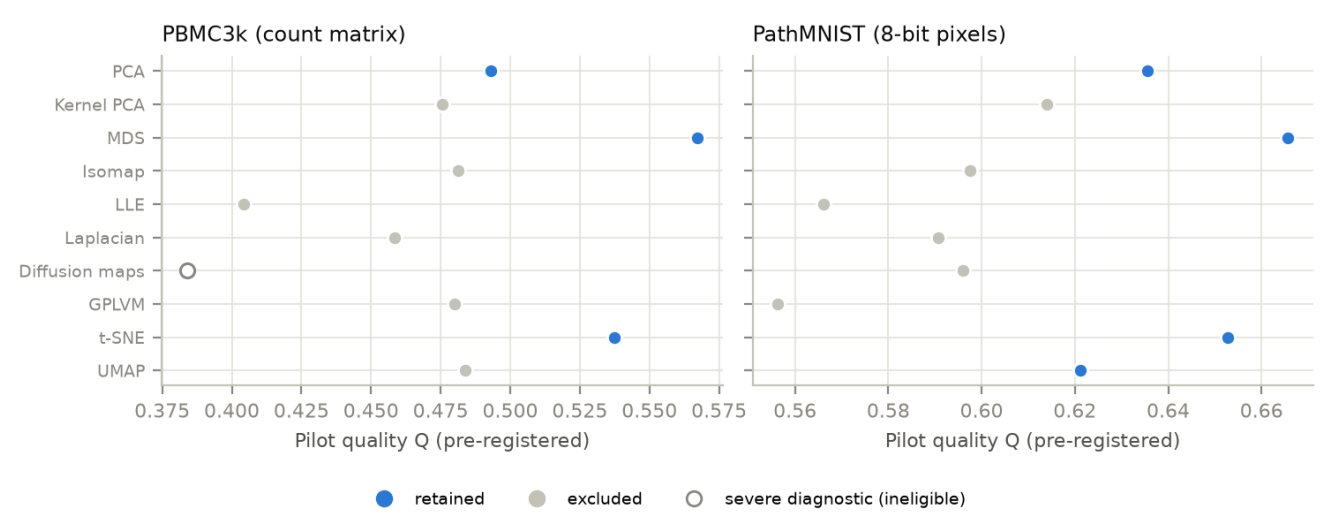


Figure 2. Pilot quality of all ten methods on each dataset. Open circles are ineligible because of a severe diagnostic. Full per-method evidence is in generated_report_1.pdf and generated_report_2.pdf.

Dataset	Shortlist (pilot Q)	Final cohort	Final T / R / G of each finalist
PBMC3k	PCA 0.493; MDS 0.567; t-SNE 0.537	2,638 (all available observations)	PCA 0.825 / 0.051 / 0.808; MDS 0.848 / 0.064 / 0.842; t-SNE 0.897 / 0.208 / 0.654
PathMNIST	PCA 0.635; MDS 0.666; t-SNE 0.653; UMAP 0.621	12,500 (per-run timeout, final-stage time budget, memory budget at n=15000)	PCA 0.843 / 0.071 / 0.858; MDS 0.850 / 0.073 / 0.955; t-SNE 0.900 / 0.186 / 0.780; UMAP 0.861 / 0.099 / 0.831

What the evidence says. On both datasets the agent kept one method strong on global structure (MDS) and one strong on local structure (t-SNE), next to the PCA baseline.

MDS had the best pilot Q on both datasets. Two components drove it: the highest global structure of all valid methods (G 0.855 on PBMC3k, 0.949 on PathMNIST) and higher subsample stability than t-SNE (S 0.756 vs 0.428; 0.813 vs 0.598). Together they outweighed t-SNE's better local recall. t-SNE preserved local neighborhoods best: on the final cohorts its recall was 0.208 vs 0.064 for MDS on PBMC3k, and 0.186 vs 0.073 on PathMNIST. Its global correlation was lower (0.654 and 0.780).

UMAP advanced on PathMNIST but not on PBMC3k, where its Q (0.484) was below both the 90% threshold and PCA. Diffusion maps on PBMC3k were ineligible: their layout collapsed and the diffusion graph was nearly disconnected (Markov gap 9e-08).

The PathMNIST final size was bounded by MDS. Its fitted time exponent was 2.30, so 15,000 images would have exceeded the budgets. The size rule needed its safety factor: MDS took 1038 s at n = 12,500, against a point projection of 757 s (1515 s with the 2× safety factor). A likely cause is the cubic eigendecomposition inside the classical start, which grows faster than the probes (at most 4,000 observations) could show; this was not timed separately.

The secondary label silhouettes support the same division of labor on PBMC3k (t-SNE highest). On PathMNIST they are negative for every finalist: pixel distances do not separate the tissue classes, a limitation of the pixel representation rather than of any one method.

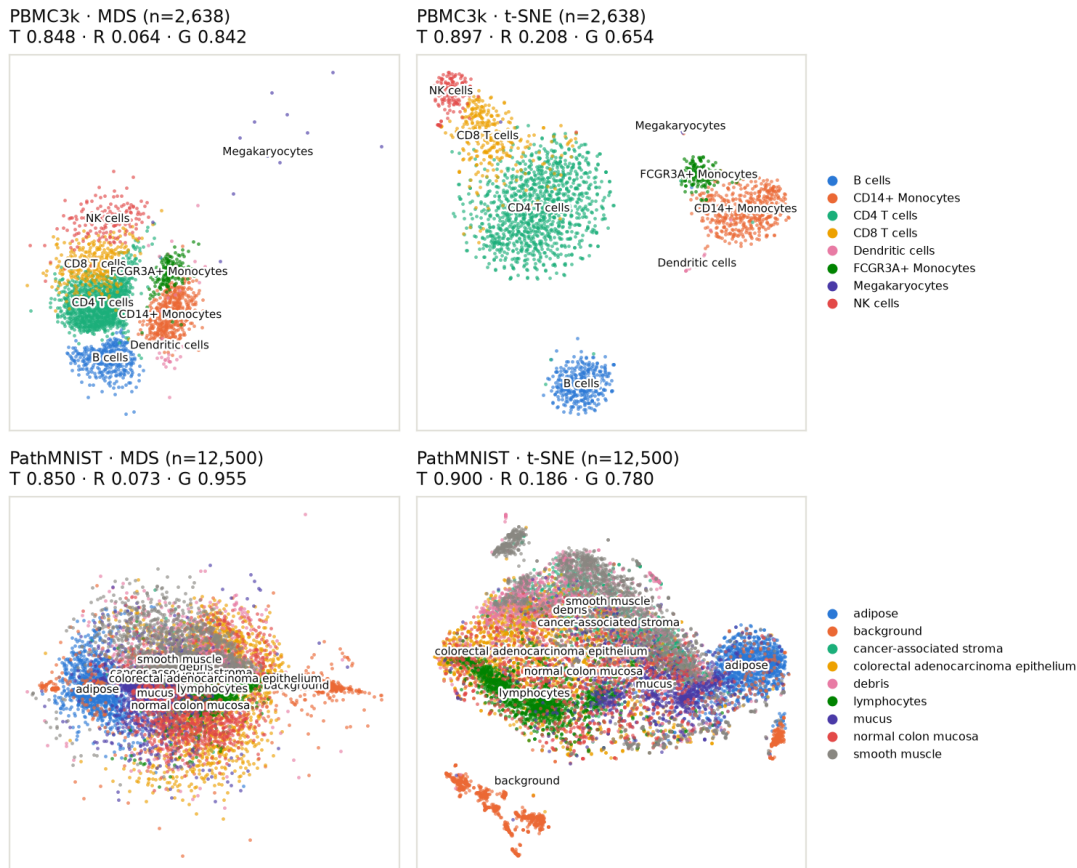


Figure 3. Final embeddings of the two nonlinear finalists common to both datasets. Class names sit at class medians. Colors are secondary annotations, and visual separation is not evidence of biological or histological truth.

5 • Robustness and reflection

Cumulative change (same v2 evidence)	PBMC3k shortlist	PathMNIST shortlist
A0 v1 rules on v2 evidence (fit-only timing)	PCA, t-SNE, MDS, Isomap	PCA, t-SNE, MDS
A1 + runtime materiality (epsilon-Pareto)	PCA, t-SNE, Isomap	PCA, t-SNE, MDS
A2 + collapse / spectral-gap / convergence checks	PCA, t-SNE, Isomap	PCA, t-SNE, MDS
A3 + global-structure component	PCA, t-SNE, MDS, Isomap	PCA, t-SNE, MDS, Isomap
A4 + subsample stability replaces seed stability	PCA, MDS, t-SNE	PCA, MDS, t-SNE, UMAP
A5 + equivalence-to-best eligibility (= primary)	PCA, MDS, t-SNE	PCA, MDS, t-SNE, UMAP

How stable are the decisions? Across ten rule variants (other weightings, $k = 50$, the v1 stability definition, ignoring runtime, and others), these are the selection frequencies:

- PBMC3k: PCA 100%, MDS 90%, Isomap 20%, GPLVM 10%, t-SNE 80%, UMAP 20%.
- PathMNIST: PCA 100%, Kernel PCA 10%, MDS 100%, Isomap 10%, t-SNE 100%, UMAP 80%.

PCA, MDS and t-SNE are therefore robust conclusions. UMAP's place on PathMNIST is moderately robust; any fourth method on PBMC3k would be fragile, and the agent kept none. The ablation shows which rule mattered: moving from v1's stability definition to subsample stability replaced Isomap by MDS on PBMC3k and by UMAP on PathMNIST.

What v1 got wrong, and what I learned.

- **Measure before you decide.** v1's runtimes were dominated by Python imports. v2 measures PCA's pilot fit at 0.004 s, while v1 recorded 1.81 s on PBMC3k (examples/v1). A strict Pareto rule then treated sub-second noise as evidence: two v1 runs on the same input, with identical qualities, retained different fourth methods. Perturbing one runtime by ± 0.2 s gave six different PathMNIST shortlists.
- **A shared metric must mean the same thing for every method.** t-SNE with PCA initialization ignores its seed. v1's "seed stability" therefore gave it a measured 1.0, the same score deterministic methods received by definition, while UMAP, MDS and GPLVM were penalized.
- **The pilot configuration is part of the evidence.** MDS started from random positions stopped at its iteration limit on PBMC3k and looked weak. Started from the classical solution, it converges (179 SMACOF iterations on the PBMC3k pilot, limit 1,000) and is the best method on both datasets. An evidence-based agent is only as fair as the candidates' default settings.
- **Diagnostics are necessary but incomplete.** The collapse check, calibrated on simulations before the real-data run, caught the collapsed diffusion maps on PBMC3k. Visually degenerate LLE and Laplacian layouts on PathMNIST

have collapse ratios of 0.076 and 0.082, just above the 0.05 threshold. They were excluded only because their Q was lower.

6 • Strengths, limitations and reproducibility

Strengths.

- All ten methods are genuine implementations and are attempted on one cohort.
- Decisions are pre-registered, machine-readable and recomputed by the validator.
- Failures and weak methods stay visible.
- One interface adapts preprocessing and cohort size to sparse counts and dense images.

Limitations.

- Q is one reasonable weighting. The robustness analysis bounds this choice but does not remove it.
- Metrics are in-cohort descriptions, not held-out estimates.
- Timing comes from single-threaded runs on one node.
- GPLVM uses a fixed epoch budget.
- PBMC annotations come from an earlier clustering, and PathMNIST pixel space does not encode histological semantics.

Reproducibility.

- The exact environment is given by explicit conda and pip locks.
- Every stage records configuration and code hashes, the git commit, host and job.
- Cohort IDs, seeds and checksums are saved.
- freeze locked the 1,751 output files before this report was rendered.
- Rerun with `sbatch scripts/slurm/run_all.sbatch config/<dataset>.yaml`.

References

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- Software: pydiffmap, GPyTorch (Gardner et al., 2018, NeurIPS), umap-learn, WeasyPrint. Exact versions: environment.lock.txt and each run manifest.

Appendix · v1 audit evidence

Source: examples/v1/ (decisions and metrics of the delivered v1 runs) and examples/v1/offline_rescore.json.

- Delivered v1 shortlists: PBMC3k PCA, t-SNE, Isomap, Laplacian; PathMNIST PCA, t-SNE, Laplacian, LLE. An earlier v1 PathMNIST run on the identical input (SHA-256 equal, all ten qualities equal to six decimals) retained kernel PCA instead of LLE.
- The offline re-scoring of the saved v1 embeddings reproduces both delivered shortlists exactly. Adding only the runtime-materiality rule removes Laplacian from PBMC3k and LLE and Laplacian from PathMNIST.

Rule and threshold definitions: docs/METHOD_NOTES.md. Full per-dataset evidence: reports/generated_report_1.pdf and reports/generated_report_2.pdf.